Morched Ammar

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PROFESSIONAL SUMMARY

Highly skilled firmware and embedded AI engineer with expertise in interrupt-driven STM32 firmware, FreeRTOS multitasking on ESP32, YOLOv8 computer vision, and PPO reinforcement learning deployed on edge hardware, proficient in C/C++, Python, and the full embedded protocol stack including CAN, OBD-II, MQTT, I2C, SPI, and BLE, with experience in automotive diagnostics, industrial IoT, and real-time computer vision, utilizing tools like STM32CubeIDE, Arduino IDE, and VS Code, and frameworks such as Flutter and Dart, with a strong background in sensor integration, actuator control, and PCB design using EasyEDA and Altium.

SKILLS

C/C++, Python, OBD-II, CAN, MQTT, I2C, SPI, BLE, Wi-Fi, TCP/IP, STM32, ESP32, Raspberry Pi, Flutter, Dart, Flask, YOLOv8, OpenCV, ONNX, Edge AI, Reinforcement Learning, PPO, Git, STM32CubeIDE, Arduino IDE, VS Code, EasyEDA, Altium, Sensor integration, Actuator control, PCB design

EXPERIENCE

Capgemini- Final year intern: mobile Dev **Feb 2025/Aout 2024**
Capgemini

- Developed a Flutter/Dart OBD-II client supporting real-time PIDs with live data streaming from ECUs, utilizing OBD-II protocol and CAN bus for automotive diagnostics.
- Designed a Flask-based ECU simulator for injecting, reading, and clearing DTC fault codes, validated against a physical ECU with high accuracy, using MQTT and TCP/IP for telemetry extraction.
- Integrated a Wi-Fi/TCP-IP telemetry extraction pipeline with ELM327 hardware dongle for real-time diagnostics and monitoring.

Sofrecom-Summer internship: embedded system **July 2024/ August 2024**
Sofrecom

- Created a REST API with 51Degrees device detection engine for identifying smartphone models and retrieving environmental metrics, including eco-rating, repairability, and climate efficiency scores, using HTTP and RESTful architecture.
- Built a Raspberry Pi backend for processing sustainability profiles and transmitting structured data to an ESP8266 over Wi-Fi, enabling near-instant results on QR code scan, utilizing I2C and SPI for sensor communication.

PROJECTS

TrafficOS — Deep RL Intersection Controller

Technologies: OpenAI Gymnasium, PPO, Stable Baselines3, Raspberry Pi 5, ONNX

Engineered a custom OpenAI Gymnasium environment modeling a 4-way intersection with an 8-dimensional observation space (per-lane vehicle density + emergency vehicle priority flags). Trained a PPO agent (Stable Baselines3) over 500,000 timesteps, achieving a 30–35% reduction in average vehicle wait time vs. fixed-timer controllers.

Padel Player Tracking — Computer Vision & IoT

Technologies: STM32F407VG, I2C, SPI, MQTT, Wi-Fi

Engineered modular embedded firmware in C on STM32F407VG managing 3 sensors (LEM LTS 6-NP, DHT11, PIR) across I2C and SPI buses with a layered driver architecture enabling sensor hot-swap without core firmware changes. Implemented MQTT-over-Wi-Fi cloud pipeline for continuous telemetry upload, enabling remote monitoring and real-time actuation commands from a cloud dashboard.

IoT Energy Monitoring System — STM32F407VG

Technologies: STM32F407VG, I2C, SPI, MQTT, Wi-Fi

Engineered modular embedded firmware in C on STM32F407VG managing 3 sensors (LEM LTS 6-NP, DHT11, PIR) across I2C and SPI buses with a layered driver architecture enabling sensor hot-swap without core firmware changes. Implemented MQTT-over-Wi-Fi cloud pipeline for continuous telemetry upload, enabling remote monitoring and real-time actuation commands from a cloud dashboard.

EDUCATION

Bachelor's degree in electronics and automation — Higher Institute of Industrial Management (2018-2022)
Bachelor's degree in computer engineering — Private higher school of engineering and technology (Sept 2022 - Oct 2025)

CERTIFICATIONS

- Nvidia: Applications of AI for Anomaly Detection (Nov2024)
- Nvidia: Applications of AI for Predictive Maintenance (Nov2024)

LANGUAGES

English (B2), French (B2)